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PI Name:	Roux, Stanley Ph.D.		
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PI Email:	sroux@austin.utexas.edu	**Fax:**	FY 512-232-9529
PI Organization Type:	UNIVERSITY	**Phone:**	512-471-4238
Organization Name:	University of Texas - Austin		
PI Address 1:	Department of Molecular Biosciences		
PI Address 2:	205 West 24th Street, STOP A6700		
PI Web Page:	http://www.sbs.utexas.edu/		
City:	Austin	**State:**	TX
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Contact Monitor:	Taylor, Elizabeth	**Contact Phone:**	650.604.1783
Contact Email:	elizabeth.taylor-23@nasa.gov		
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Ed. NOTE (June 2015): Continuation for SporeSat 2 of "Sporesat Mission: Investigating biophysical mechanisms of plant gravisensing using a lab-on-a-chip approach"--grant NNX11AF48A.
SporeSat is a Small Complete Mission based on a Mission of Opportunity-2 nanosatellite designed for real time measurement of calcium signaling and maintenance of variable gravity treatments during flight. The impetus for SporeSat is to gain a deeper knowledge of the molecular and biophysical mechanisms for plant gravity sensing by studying the activation of cell gravity-sensing and electrophysiological signaling in the single-cell model system, Ceratopteris richardii, using a novel ‘lab-on-a-chip’ microsensor technology platform called a CEL-C advanced bioCD.

In 2013, SporeSat was adapted into a 2 flight mission strategy with flight 1 focused on the technology objectives and flight 2 focused on the scientific objectives. In April 2014, SporeSat-1 was deployed from a SpaceX Falcon 9 launched from Cape Canaveral as part of CRS-3. SporeSat-1 was successfully commanded and science data downloaded.

In late 2012, the original SporeSat Principal Investigator (PI), Dr. Marshall Porterfield, was tapped as the director of NASA’s Space Life and Physical Sciences Research and Applications Division. Because SporeSat falls under this division at NASA, due to conflict of interest, Dr. Porterfield was replaced as PI with his close research collaborator and fellow expert in physiological sensing and biosensor technology, Dr. Jenna Rickus. Due to the increased emphasis on the Science Objectives for SporeSat-2, Drs. Stan Roux (original co-Investigator) and Rickus will serve as Co-Principal Investigators for SporeSat-2. Dr. Rickus will lead all aspects of the bioCD design, performance, and production at Purdue and Dr. Roux (at University of Texas Austin) will lead all biological aspects of the project particularly that related to experimental design and interpretation of the experimental data as well as setting the science requirements for the SporeSat-2 mission.

At the time of loading for SporeSat-1 the Hn-n Ceratopteris richardii spores were not available; therefore Rn5 spores from the University of Texas (UT) were used. In the SporeSat 2 mission we will use the now available Hn-n spore strain that was used in all previous calcium current studies prior to SporeSat. Using the Hn-n from UT Austin and the upgraded SporeSat-2 hardware we will record ground based calcium currents in stationary positions (variable = angle off gravity) and under rotation (variable = artificial gravity magnitude). In addition, as mutant spores become available we will integrate the mutants into the experiments.

The SporeSat-2 experimental paradigm and geometry is different from any previous experimental set ups. As in any biosensor experiment, the Ca2+ ISEs are sampling at a particular location in space and time, a dynamic calcium current that exists across the 3D surface of the fern spore. To assist in development of the biological model based on sensor recordings, we have developed a 3D simulation model that predicts the 3D calcium concentration profiles in the SporeSat wells as well as the sensor recordings over time based on the biological flux from the fern spore. We will therefore be able to compare our experimental data with our predictions for different conditions (angle off gravity, under rotation) as well as for different calcium mutants.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Astronaut health in long term space missions is likely to be dependent on and benefit from plants. On Earth we depend on plants for food, clean air, and waste treatment. The same will be true in space. If we as humans are to achieve the kind of space exploration and habitation that we all imagine and dream of, we are going to need plants up there with us. Yet on the International Space Station (ISS), gravity is less than one one-thousandth of the gravity here on Earth. On the Moon, it is about 17% of Earth's gravity. In addition, there are astonishing similarities in signaling within cells across species. As in plants, calcium plays important roles in human cell development, function, and dysfunction. Understanding gravity sensing in plant cells is likely to give clues to gravity sensing in other organisms including humans. In addition, the technology that we developed for these experiments establishes a platform to ask other questions regarding human cell responses on Earth and in space. We are already looking at how we can adapt the technology to study pancreatic beta cells and diabetes.

Light Initiation of Spore Germination
We have determined that 1 hour of light is sufficient to initiate the germination of spores and allow the gravity-directed polarization process to proceed through 24 hours. This assures that the battery power planned for the light treatment in the SporeSat mission will easily suffice. Spores require an additional light exposure between 28 and 40 hours after the first light treatment to complete germination, but the calcium ion transport investigated in the SporeSat mission occurs within the first 24 hours of development. Phytochrome is the photoreceptor for the first light treatment, and we have discovered that transcripts for all three of the Ceratopteris phytochrome genes are present in spores after one hour of light exposure. These new data provide important evidence that the photoreceptor and irradiance needed to initiate the calcium current are present during the first hour of light treatment planned.

Spore and Media Loading

Throughout the SporeSat 1 project and at the beginning of SporeSat-2, loading the BioCD with spores and agar germination media was completely manual. Capillaries pulled to 5-10 uM tip diameter were used for spore loading. Spores were allowed to dry on filter paper. Using a stereo microscope with 10X-40X total magnification, a capillary was used by hand to pick up one spore via electrostatic attraction of the spore to the capillary side. The spore was placed on top of a well and the spore was gently pushed in the well using the side of the capillary. This process was repeated until all wells were loaded with one spore, a 6-8 hour process. After all wells were loaded with spores a 2 µl micropipette was used to apply 0.5-1.5 µl of 2% agar germination media to each of the wells. This process was unsatisfactory in that it often resulted in breaking open spores during loading, the complete dry-out of the previously hydrated spores, and incomplete filling of most wells due to unavoidable bubbles in the agar media. Early in phase one of SporeSat-2 we found that this drying out of the spores after hydration results in inhibition or significant delay in germination. To solve this loading problem scientists at the Ames Research Center (ARC) implemented the use of a capillary pipette and micromanipulator for spore loading and adapted an off the shelf 3D printer to automatically load agar germination media into the wells. These improvements resulted in spore loading time of 2-4 hours, and complete filling of greater than 50% of wells without bubbles. We have determined that complete loading of the well with germination media without bubbles is necessary for resolution of the BioCD signal. We also altered the experimental procedure to include a dark stasis period of 3-5 days for the Bio CD after it was loaded.

Task Progress:

Germination Media Calcium Content

Using data obtained from the self-referencing probe studies of spore calcium transport, our collaborators generated a theoretical model of calcium concentration differential expected in the SporeSat BioCd electrode configuration and size to predict the calcium differential that would exist in the BiCd. This 3D model predicts that the maximum calcium differential detected by the BioCd to be around 5 µM, which corresponds to approximately 5 mV signal on the BioCd at 1 g. In initial tests we were able to detect small signals in wells containing spores that appeared different from the negative control, but there was considerable fluctuation and high deviation of the signal. Because of this high noise and small signal, we began to seek ways to improve our signal-to-noise ratio.

One method of improvement proposed was to reduce the background concentration of Ca2+ in the medium. The Sigma-Aldrich product Bactoagar was used in germination media for SporeSat-1 and early tests in Phase one of SporeSat-2. According to the manufacturers' specifications, this product contains 850 µM Ca2+. This background amount of Ca2+ is well above the spore generated calcium differential predicted, and would at least partially mask any biologically generated differential from detection. We found several additional agar products with lower reported Ca2+ content, and used these products in spore germination media to test germination rate. We initially tested one product, Type A agar, for its ability to support spore germination. We found that the percent of germinating spores was reduced, and the development of gametophytes was altered compared to other gelling agents. Type A agar was omitted from further studies. The agar manufacturer's specifications did not list specific calcium content by lot for any of the products, and there were no data available on the calcium content of Ultrapure Agarose. In order to determine the exact calcium content of the spore germination media, we prepared media for Inductively Coupled Plasma Mass Spectrometry (ICP-MS) analysis at the Quadrupole ICP-MS Lab at the University of Texas at Austin. In order to perform this analysis, acid digestion of media samples was necessary. We determined that the agar media with the lowest background calcium concentration that supported both normal spore germination and strong electrochemical signals from spores was Noble agar, which had a background [Ca2+] of ~ 280 µM, and this became the agar of choice for the successful test run we completed in April (Zero Day Test 12). Although "scrubbing" agarose preparations with a BAPTA chelating product was able to lower their [Ca2+] further, the BAPTA-treated agarose preparations were unable to support normal spore germination, so we could not use "scrubbed" agarose preparations, even though they offered a lower background [Ca2+].

Spore Load Improvement Testing

With the implementation of spore and media loading improvements we were able to achieve 50-70% successful loading of wells on a BioCd. The major limitation to complete loading of agar was the high hydrophobicity of the ion selective membrane on the BioCd. To improve loading success we tested the addition of surfactant to the spore germination media. The addition of 0.05% (0.8mM) triton X-100 to spore germination media for the first 24 hours of development followed by removal of the surfactant for the remainder of development did not alter the germination timing or percent of germinating spores. When added to germination media this concentration of surfactant improved the successful loading rate to 83% in tests without spores. We found, however, that continuous exposure to 0.05% triton X-100 throughout spore germination results in lysis of all spores before normal germination. This spore lysis occurred in a much lower percent of the population if lower concentration of triton X-100 (0.01% or 0.005%) was added; however, these lower concentrations of triton X-100 did not result in the same improvement in agar loading. Further tests convinced us to load the spores into the BioCd without triton X-100.

Success in Detecting Gravity-Directed Electrical Signals from Spores

Thanks to the reduced background Ca2+ in the agar, improvements in spore loading, and an improvement made to the signal detection software, we have developed an effective BioCd. Dr. Rickus was able to detect statistically significant electrical signal from spore cells that increases with g-force, and is different from negative controls in Zero Day test #12. This success replicated results observed in prior Zero Day tests, but with a dramatic reduction in noise and increase in signal stability. These results indicate that spore, spore loading, and instrument conditions have been found that reproducibly confirm that the bioCD can record significant electrical signals in the spore that are induced by gravity. The likelihood that these signals are due to changes in calcium transport is very high, and phase 2 will include experiments that test this likelihood.

Mutant Plant Development

During the first two years of Phase A of the SporeSat grant much of the research effort was focused on extensive hardware development, perfecting the batteries, master PCB, lighting, and accompanying electronics of the BioCD that would fly the Ceratopteris spores, and this research was done mainly at ARC and Purdue. These hardware efforts had progressed significantly by 2014, when an initial flight was carried out, and only then did research to generate Ceratopteris mutants begin. This mutant work was carried out at the University of Texas, beginning in the Fall of 2014. At this time only two well described methods for generating stable transformants in Ceratopteris were available (Muthukumar et al., 2013; Plackett et al., 2014), and neither of these method papers described the successful suppression or overexpression of any endogenous Ceratopteris gene. Perfecting transformation methods was therefore challenging, but, as described below, ultimately successful.

The original SporeSat proposal included the objective of observing the calcium current response of knockout mutant spores for the Ceratopteris Ca2+ ATPase (CrACA1) and Annexin (AnnCr1 and AnnCr2) genes. The CrACA1 gene product is very likely responsible for Ca2+ efflux from germinating spores (Bushart et al., 2014). As noted in the original proposal the annexin proteins have an important regulatory role in Ca2+ -mediated signaling (Laohavisit and Davies, 2011), including polar development of plant cells and calcium influx (Wang et al., 2010; Tang et al., 2014; Zhu et al., 2014). When we discovered that AnnCr2 is the annexin transcript most abundant during spore germination, and that transcripts for AnnCr1 are not present in spores throughout the period of polarity fixation, we focused our annexin mutational studies on modifying AnnCr2 rather than AnnCr1.

At this time we have generated mutant sporophyte plants that possess short hairpin constructs designed to reduce or suppress the expression of the CrACA1 and AnnCr2 genes. In addition to these mutant plants, we have also generated mutant sporophyte plants that possess short hairpin constructs designed to suppress the expression of genes that encode a Mechanosensitive Channel of Small Conductance (CrMscS) and the ubiquitous calcium regulatory protein Calmodulin (CrCaM). We have also generated sporophyte plants that are up-regulated in the expression of the AnnCr2 and CrMscS genes.

Mechanosensitive channels are multi-pass, integral membrane proteins that form channels in membrane lipid bilayers. These channels open in response to pressure applied to the lipid bilayer (Malcolm et al., 2015). Mechanosensitive channels have long been proposed as the gravity sensing receptor in plant cells through the hypothesis that the mass of the cytoplasm or specific cellular constituents would cause pressure to be applied to the membrane locally and lead to opening of these channels (Iida et al., 2014). The opening of mechanosensitive channels in plants and other organisms allows the transport of calcium into cells (Kurusu et al., 2013). At the time the original SporeSat proposal was submitted we did not have gene sequence for any mechanosensitive channels in Ceratopteris richardii. We have since found several Ceratopteris genes encoding MscS type mechanosensitive channels. We have only found expression of one member of this gene family in spores, therefore this is the gene family member we have targeted for in plant genetic mutation analysis.

We are currently in the process of evaluating the transcript abundance of the target gene in each of the lines. The creation of knockout and overexpression mutations in plants occurs through random insertion of the DNA construct into the plant's genome. Because of the random location of insertion, in some cases the introduced product does not result in the expected change in transcription. Also, the location of the insertion and not the inserted product may result in phenotypical alterations to the plant. For these reasons we will maintain and test at least three different independent lines with each of the genetic mutations.

We have currently verified three plants with significant increase in transcript abundance of AnnCr2 and three plants with significant decrease in transcript abundance of AnnCr2. We have also verified that one plant has significant increase in transcript abundance of CrMscS. We are collecting spores from all of these plants that will be tested for appropriate transcriptional changes and possible gravity-related differences.

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Bibliography Type:	Description: (Last Updated: 10/16/2023)
Abstracts for Journals and Proceedings	Salmi M, Malcolm H, Cantero-Garcia A, Clark GB, Roux SJ. "Characterization of mechanosensitive channels from a graviresponsive fern spore." Presented at the 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 12-15, 2015. Meeting Program. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 12-15, 2015. , Nov-2015
Articles in Peer-reviewed Journals	Cannon AE, Salmi ML, Clark G, Roux S. "New insights in plant biology gained from research in space." Gravit Space Res. 2015 Dec;3(2):3-19. Review. http://gravitationalandspacebiology.org/index.php/journal/article/view/692/726 , Dec-2015
Journal/Magazine covers	Cannon AE, Salmi ML, Clark G, Roux S. "Cover in journal Gravitational and Space Research for article, 'New insights in plant biology gained from research in space.' " Gravit Space Res. 2015 Dec;3(2):3-19. Review. http://gravitationalandspacebiology.org/index.php/journal , Dec-2015